

ROHAN KUMAR

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PROFESSIONAL SUMMARY

Senior Risk and Compliance Analytics professional with 16+ years of total experience, including 8+ years across global banking institutions including SMBC, Deutsche Bank, HSBC, and Santander. Experienced in trade surveillance, credit risk modeling, regulatory analytics, and model governance frameworks such as SR 11-7 and CCAR. Proven track record of developing and enhancing Market Surveillance models for Fixed Income and FX markets in alignment with SEC, FINRA, and global banking standards. Recognized for translating complex analytics into compliant, auditable solutions and building effective partnerships across Trading, Technology, and Risk functions.

PROFESSIONAL EXPERIENCE

Madison-Davis / SMBC Compliance, Jersey City, NJ

Compliance Analytics Consultant | Jun 2025 - Dec 2025

- Developed and enhanced in-house Market Surveillance models to detect potential market manipulation and policy-breach behaviors across Fixed Income and FX asset classes, improving detection precision while reducing false positives by 20%.
- Built analytical workflows using Azure Databricks (Python/SQL) to automate surveillance model inputs and alert reviews, cutting manual processing time on a daily basis by 20-30%.
- Performed data reconciliation, pattern detection, and anomaly analysis using structured and unstructured datasets to strengthen surveillance coverage.
- Conducted annual threshold tuning and back-testing (BTL) exercises for in-house surveillance models to improve detection sensitivity and reduce false positives.
- Collaborated with Compliance, Technology, and Surveillance SMEs to ensure model logic aligns with SEC, FINRA, and internal SMBC policies.
- Prepared methodology documentation, model change artifacts, and test evidence in alignment with SR 11-7 model risk standards.
- Performed code versioning and model enhancements using GitHub; documented alert logic, UAT/production run procedures in Confluence.

PHD/M.S. Computer Science - Independent Analytics Research, Jersey City, NJ

Self-Directed | Analytics Research & Applied ML Development | Jan 2024 - Present

- Designed and developed a Trade Surveillance Detection System (Python, OOP) integrating real-time market data via Yahoo Finance and CoinGecko APIs; published ML-based anomaly detection research on SSRN (Abstract ID 6576018) covering Fixed Income and crypto market surveillance.
- Built an end-to-end Customer Support Operations Dashboard in Power BI using Databricks SQL; designed DAX measures, KPI visuals, and executive-level data storytelling layouts.
- Completed advanced coursework in Machine Learning, Data Engineering, and Algorithms as part of M.S. Computer Science program (Monroe University); applied regression modeling and Jupyter-based analysis to real-world datasets.
- Maintained subject matter expertise in financial risk and regulatory compliance through continued engagement with industry literature on BCBS 239, SR 11-7, CCAR/PPNR, and market surveillance frameworks.

Deutsche Bank, New York, NY

Assistant Vice President - Senior Risk Methodology Specialist | Aug 2022 - Dec 2023

- Led strategic analytics for corporate banking portfolios supporting CCAR stress testing and IRRBB, managing financial models with \$5B+ in revenue and balance sheet exposures.
- Revamped and calibrated PPNR models, leading to a 15% reduction in forecasting errors and increased model accuracy.
- Supported CCAR stress testing 2023 and MRM validation activities for Corporate Trust, Institutional Cash Management, and Corporate Cash Management portfolios.
- Monitored CCAR/PPNR models and ensured compliance with model risk management standards per SR 11-7 and internal governance policies.
- Supported IRRBB analytics by analyzing interest rate sensitivity, balance sheet repricing gaps, and scenario impacts under regulatory stress environments.
- Collaborated with stakeholders across teams to ensure compliance with bank standards and guidelines, resulting in a 30% faster model review cycles, reporting turnaround.

Primesoft Consulting Services (Santander), Jersey City, NJ*Credit Risk Modeler (Contract) | Apr 2020 - Aug 2022*

- Developed credit models including risk rating, scorecard, and CCAR loss estimation models leveraging PD/EAD/LGD frameworks, stress testing, and prepayment models.
- Led NDOD (IFRS 9) validation analysis for retail portfolio using SAS, achieving compliance with model development standards.
- Improved data accuracy and efficiency of risk reporting using SQL and SAS for extraction, transformation, and analysis of large, structured datasets.
- Prepared model development documentation for CECL and IFRS 9 compliance, ensuring 98% adherence to established standards.

Huxley / Allegis Global Solution (HSBC), New York, NY*Risk Management Specialist (Contract) | Jun 2018 - Nov 2019*

- Managed fraud risk models and related strategies; performed independent model reviews and oversaw vendor model management.
- Executed fraud model monitoring using SAS Enterprise Guide, ensuring 99% compliance with bank standards; conducted annual vendor model reviews leading to 15% improvement in model efficacy.
- Collaborated with IT and model implementation groups for UAT and BRD documentation, achieving 95% accuracy in documentation completeness.

CRISIL / S&P Global, New York, NY*Quantitative Analyst | Sep 2015 - Nov 2017*

- Supported CCAR stress testing, time series regression model development, and validation of fraud vendor models and PPNR models.
- Validated models for large investment banks (Citi, BofA, BNP), improving model governance by implementing standardized validation and monitoring processes.
- Developed time series regression models and created SAS datasets, reducing data processing time by 20%.

Kodukulla and Associates Inc., Chicago, IL*Research Analyst | Jan 2015 - May 2015*

- Monte Carlo Simulation and DCF analysis Applied to assess project NPV distributions and risk ranges across multiple industries; developed client training materials on quantitative risk methodology, improving estimation accuracy by 10%.

EARLIER CAREER - ENGINEERING & TECHNOLOGY

HCL Technologies - Lead Engineer, Pune, India (May 2012 - Jul 2013)**Tata Motors** - Engineer, Pune, India (Jan 2009 - Nov 2011)**Tech Mahindra** - Developer/Analyst, Hyderabad, India (Dec 2007 - Dec 2008)**Action Construction Equipment / TAG Design** - Design Engineer, India (Sep 2005 - Dec 2007)**EDUCATION**

Harrisburg University of Science and Technology, PA – PHD Computational Science and Engineering (Expected 2029)**Monroe University, New Rochelle, NY** - M.S. Computer Science, GPA: 4.0**Harrisburg University of Science and Technology, PA** - M.S. Analytics, GPA: 3.9**Goldey-Beacom College, Wilmington, DE** - MBA, Financial Management, GPA: 3.95**Illinois Institute of Technology, Chicago, IL** - M.S., Finance - Risk Management & Corporate Finance**Manipal Institute of Technology, India** - B.E., Mechanical Engineering**CERTIFICATIONS**

- Databricks Certified Data Engineer Associate
- Equity Essentials, Bloomberg Essentials Training Program, Chicago, IL

TECHNICAL SKILLS

Languages & Analytics: Python, SAS, R, SQL, PostgreSQL, kdb+/q, NoSQL, SQLite**Big Data / Cloud:** Azure Databricks, PySpark, Azure, GCP, AWS**ML & Libraries:** TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, SpaCy

Visualization & BI: Tableau, Power BI, Matplotlib, ggplot

Tools: JIRA, Confluence, MS Project, GitHub, Visual Studio Code, SQL Server

Deep Learning: CNN, RNN, GAN, RAG, Deep Learning Frameworks & Techniques

CORE COMPETENCIES

Trade Compliance & Market Surveillance, Fixed Income & FX Risk, Regulatory Compliance (SEC, FINRA, SR 11-7),
Model Development, Validation & Governance, CCAR Stress Testing, IRRBB, Credit Risk (PD/EAD/LGD),
Predictive Modeling & Machine Learning, Time Series Forecasting, Statistical Analysis,
Stakeholder Management, Cross-Functional Collaboration, Regulatory Documentation

Note: I am currently an F-1 PhD student in Computational Science and Engineering. I am eligible to obtain CPT authorization through my university for a role related to my doctoral program. CPT does not require employer visa sponsorship.